Calculus II Lesson 25 Conics Sections - Parabola

Equations of Parabola:

1) Parabola opens up: $(x-h)^2 = 4p(y-k)$

vertex $=(h,k)$; focus $=(h,k+p)$; directrix: $y = k - p$

2) Parabola opens down: $(x-h)^2 = -4p(y-k)$

vertex $=(h,k)$; focus $=(h,k-p)$; directrix: $y = k + p$

3) Parabola opens right: $(y-k)^2 = 4p(x-h)$

vertex $=(h,k)$; focus $=(h+p,k)$; directrix: $x = h-p$

4) Parabola opens left: $(y-k)^2 = -4p(x-h)$

vertex $=(h,k)$; focus $=(h-p,k)$; directrix: $x = h+p$

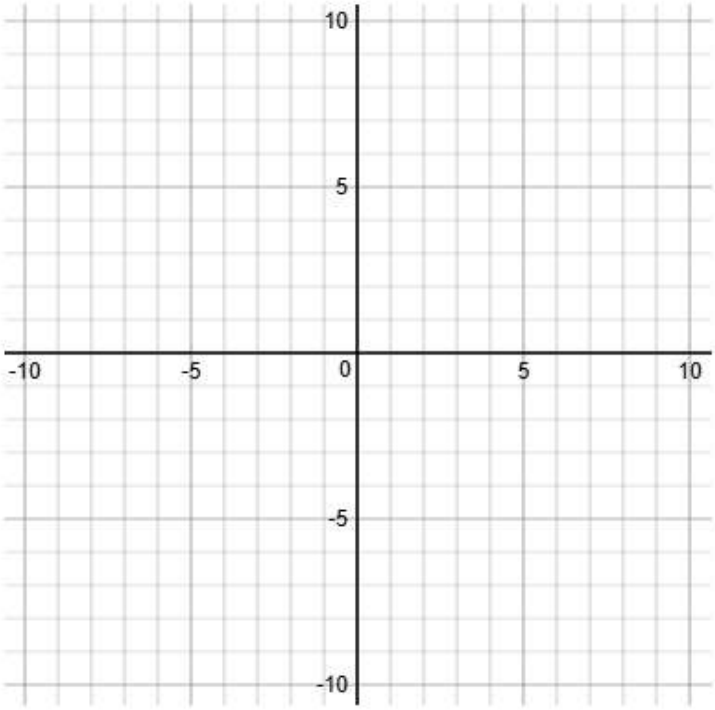
1) $y^2 = 10x$

a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph $y^2 = 10x$



2) $y^2 - 6y = 10x$

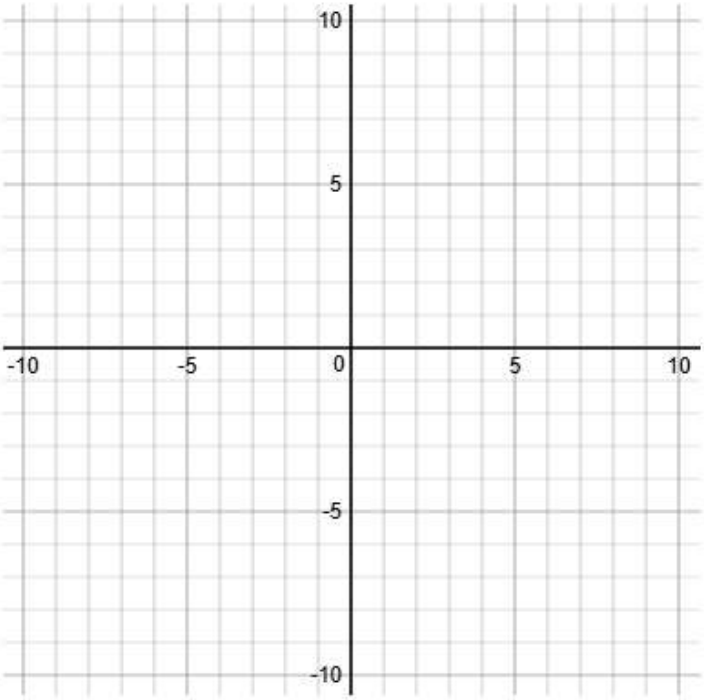
a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph $y^2 - 6y = 10x$

Hint: Use the online 2D Calculator.



3) $x^2 + 8x + 8y - 10 = 0$

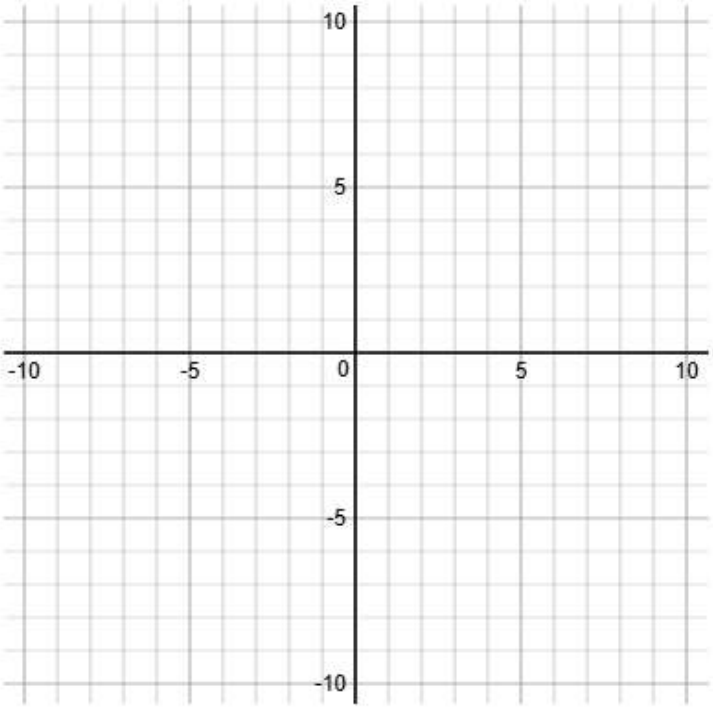
a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph $x^2 + 8x + 8y - 10 = 0$

Hint: Use the online 2D Calculator.



4) Find an equation of the parabola that has the following vertex and focus:
vertex: (4, 3) ; focus: (1, 3)

a) p = distance from vertex to directrix = distance from vertex to focus = _____

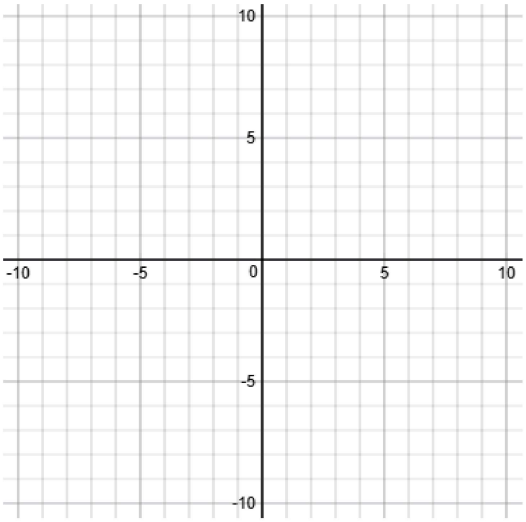
b) Equation of parabola:_____ (multiple choice)

A) $(y-3)^2 = -6(x-4)$ B) $(y-3)^2 = -8(x-4)$

C) $(y-3)^2 = -12(x-4)$ D) $(y-3)^2 = -14(x-4)$

c) Graph equation of parabola.

Hint: Use the online 2D Calculator.



5) Find an equation of the parabola that has the following vertex and directrix:
vertex: (4, 3) ; directrix: $x = 2$

a) p = distance from vertex to directrix = _____

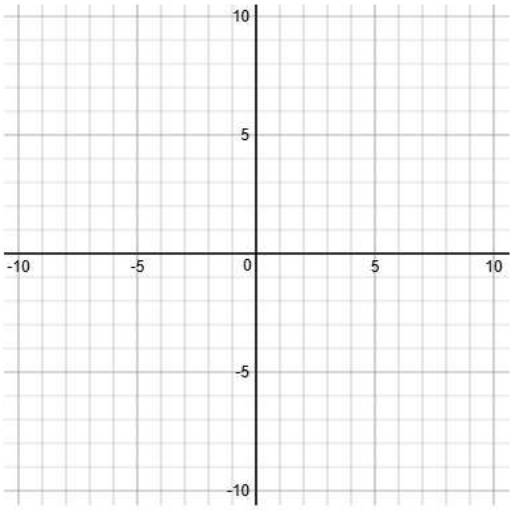
b) Equation of parabola:_____ (multiple choice)

A) $(y-3)^2 = 2(x-4)$ B) $(y-3)^2 = 3(x-4)$

C) $(y-3)^2 = 8(x-4)$ D) $(y-3)^2 = 5(x-4)$

c) Graph equation of parabola.

Hint: Use the online 2D Calculator.



6) $x^2 - 4x + 4y - 12 = 0$

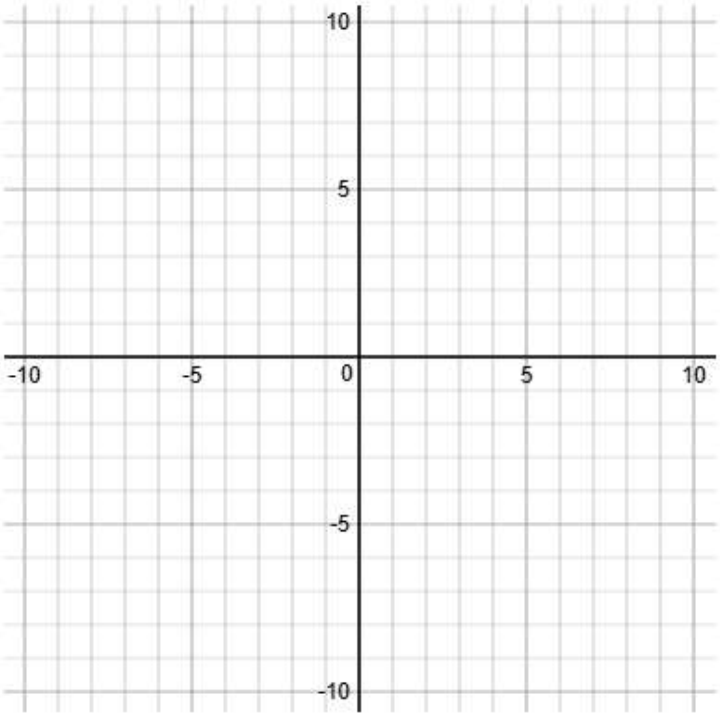
a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Sketch graph:

Hint: Use the online 2D Calculator.



7) $y^2 - 4x + 2y - 8 = 0$

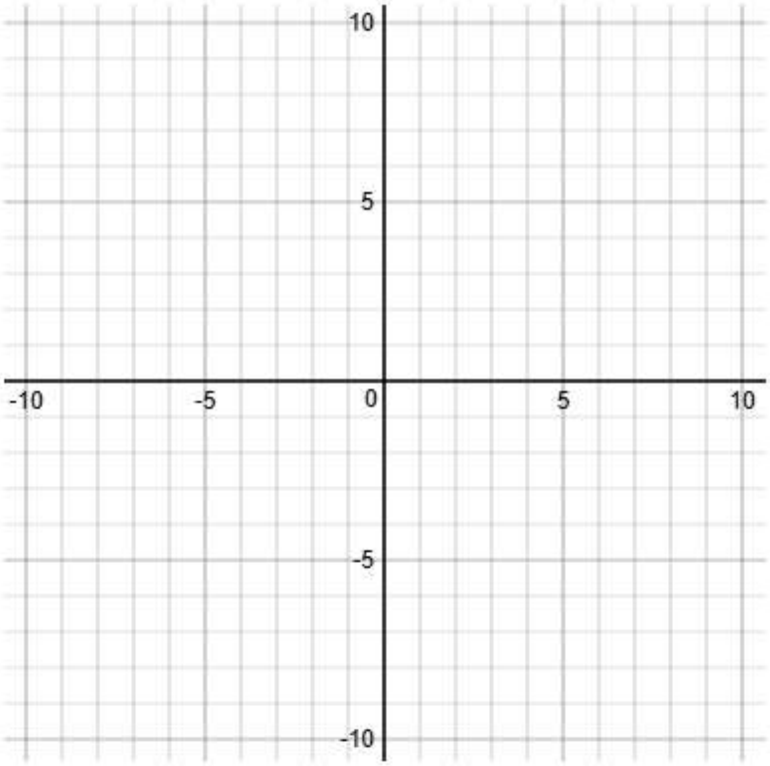
a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Sketch graph:

Hint: Use the online 2D Calculator.



8) Find an equation of the parabola that has the following vertex and focus:

vertex: $(4, 5)$; focus: $(4, -3)$

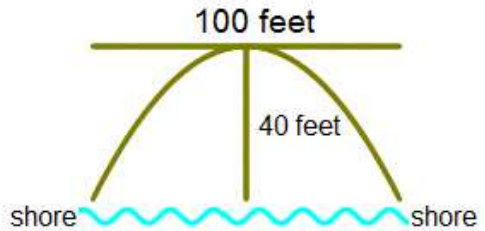
Equation of parabola:_____

9) Find an equation of the parabola that has the following vertex and directrix:

vertex: $(6, 3)$; directrix: $y = -2$

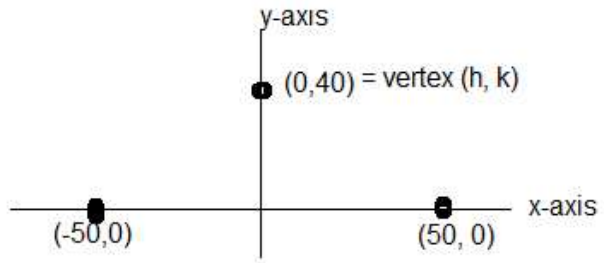
Equation of parabola:_____

10) A bridge is designed in the shape of a parabolic arch. The road over the bridge is 100 feet long and the maximum height is 40 feet.



A) Write an equation for the parabolic arch.

Hint: (h, k) = (0, 40); to find *p* use (x, y) = (50, 0).



B) If you are at a location 5 feet from the shore, how far above the water is the bridge?

Hint: Let x = 45, and find y.

11) A baseball was hit 3 feet from the ground level with speed of 120 feet/sec and angle of elevation is $\theta = 30^\circ = \pi/6$ rad.

Equation for the ball's path:

$$y = h + v_0 \sin\theta \left[\frac{x}{v_0 \cos\theta}\right] - \frac{1}{2}\cdot g\left[\frac{x}{v_0\cos\theta}\right]^2; \qquad g = \text{ Earth's gravity} = 32 \text{ ft/sec}^2$$

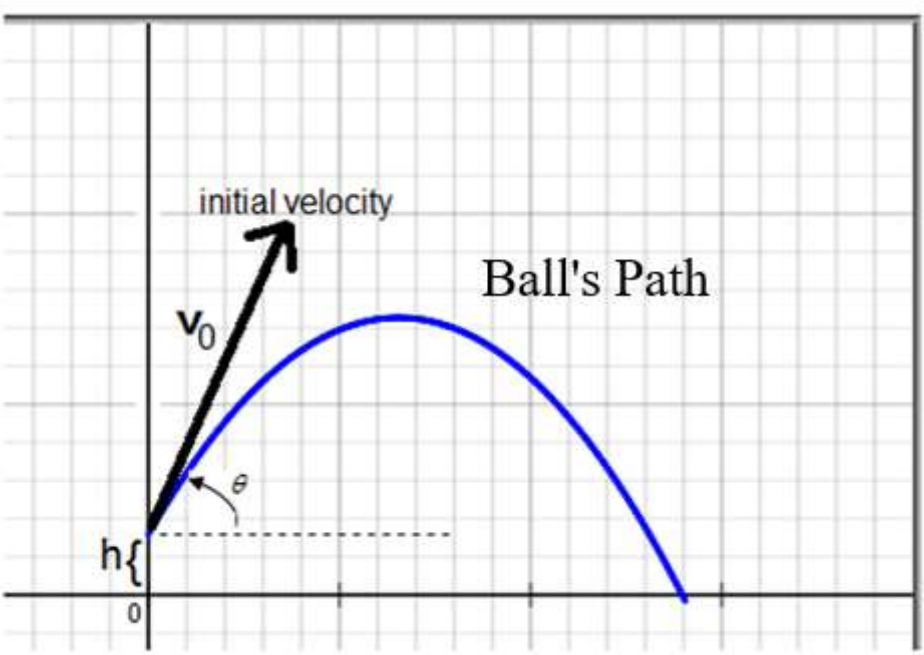
$$y = 3 + 120\sin(\pi/6)\left[\frac{x}{120\cos(\pi/6)}\right] - \frac{1}{2}\cdot 32\left[\frac{x}{120\cos(\pi/6)}\right]^2$$

$$y = 3 + 60\left[\frac{x}{60\sqrt{3}}\right] - \frac{1}{2}\cdot 32\left[\frac{x}{60\sqrt{3}}\right]^2 = 3 + \frac{1}{\sqrt{3}}x - \frac{16}{\left(60\sqrt{3}\right)^2}x^2$$

$$y = 3 + 0.57735x - 0.00148148148x^2$$

Find the maximum height reached by the ball.

a) $y' =$ _____?_____

b) Set $y' = 0$ and solve for x. $x =$ _____?_____

c) Ball reached maximum height at the point: $x =$ _____?_____ and $y =$ _____?_____